

CTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2828 ±0.0037	0.4069 ±0.0052	0.3497 ±0.0157	0.3142 ±0.00334
Cramér $\ell_2^2 \downarrow$	0.02714 ±0.000999	0.02146 ±0.000574	0.03721 ±0.00168	0.02817 ±0.00116
CF-$\ell_2^2/\omega^2 \downarrow$	0.02108 ±0.000167	0.01707 ±0.000582	0.03091 ±0.000955	0.02223 ±0.000414

QTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2002 ±0.011	0.3983 ±0.0228	0.365 ±0.0417	0.2474 ±0.0203
Cramér $\ell_2^2 \downarrow$	0.009578 ±0.00115	0.01697 ±0.00317	0.03881 ±0.00965	0.01364 ±0.00353
CF-$\ell_2^2/\omega^2 \downarrow$	0.009253 ±0.00119	0.01652 ±0.00325	0.0385 ±0.00963	0.01307 ±0.0032

ϕ TD Dirac

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1324 ±0.0152	0.4317 ±0.0514	0.2295 ±0.0464	0.1553 ±0.00698
Cramér $\ell_2^2 \downarrow$	0.004635 ±0.00128	0.01392 ±0.0048	0.01026 ±0.00135	0.003766 ±0.000377
CF-$\ell_2^2/\omega^2 \downarrow$	0.003932 ±0.00127	0.01165 ±0.00493	0.009165 ±0.000958	0.00305 ±0.000405

ϕ TD Cat

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2886 ±0.00634	0.4218 ±0.0478	0.3471 ±0.0389	0.3172 ±0.00236
Cramér $\ell_2^2 \downarrow$	0.02829 ±0.00188	0.02243 ±0.00343	0.0353 ±0.000891	0.0281 ±0.000629
CF-$\ell_2^2/\omega^2 \downarrow$	0.0221 ±0.00147	0.01837 ±0.00214	0.02995 ±0.00137	0.02257 ±0.000721

ϕ TD Quant

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1438 ±0.0179	0.3833 ±0.0458	0.1894 ±0.0131	0.1675 ±0.025
Cramér $\ell_2^2 \downarrow$	0.005254 ±0.0015	0.01022 ±0.00275	0.008621 ±0.00056	0.004322 ±0.00157
CF-$\ell_2^2/\omega^2 \downarrow$	0.004625 ±0.00139	0.009352 ±0.00291	0.008134 ±0.00056	0.003509 ±0.00126